University of Exeter

MTH2010

Groups, Rings and Fields

On Campus Exam	Closed Book
Additional Materials: None	**Permitted Materials:** Approved Non-Programmable Scientific Calculators Permitted (Green)

Guidelines:

-The **duration** of this paper is 2 hours
-Word Count: No Word Count
-Write all answers in the answer booklet provided.
-Do not open the exam paper until instructed by the Invigilator.
-Use only black pen or pencil.
-Mobile and electronic devices are not allowed.
-Do not communicate with other candidates during the exam.

Exam Instructions:

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each). Marks shown in questions are merely a guideline.

SECTION A

All rings are commutative with an identity element 1 **and** $1 \neq 0$**.**

1. (a) Let G be a group.

   (i) Let H and K be subgroups of G. Show that the intersection $H \cap K$ of H and K is a subgroup of G. (8)

   (ii) State what it means for a subgroup of G to be a normal subgroup. (5)

   (iii) If H and K are normal subgroups of G, is the intersection $H \cap K$ a normal subgroup of G? Justify your answer. (8)

   (b) Let D_8 be the dihedral group with 8 elements.

   (i) Find the conjugacy classes of D_8, listing the elements of each conjugacy class. (10)

   (ii) Write down the class equation for the group D_8 and find the centre $Z(D_8)$ of D_8. (5)

   (c) (i) Let R be a ring. State what it means for a non zero element a of R to be a zero divisor. State what it means for R to be an integral domain. (6)

   (ii) Let $\mathbf{Q}[X]$ be the ring of polynomials with coefficients in the field $\mathbf{Q}$ of rational numbers. Prove that the ring $\mathbf{Q}[X]$ is an integral domain. (8)

[**50**]

SECTION B

All rings are commutative with an identity element 1 **and** $1 \neq 0$**.**

2. Let G be a **finite** group. Consider the group action of G on itself by conjugation, which is defined by

$$G \times G \to G$$

$$(h, g) \mapsto hgh^{-1}.$$

(a) For an element $g \in G$, define what the orbit $\text{orb}(g)$ and what the stabiliser $\text{Stab}(g)$ of g are for the conjugation action. Explain what the orbit-stabiliser theorem states in this case. (6)

(b) Prove that the cardinality of $\text{orb}(g)$ is 1 if and only if g is an element of the centre $Z(G)$ of G. (6)

(c) Explain why two orbits $\text{orb}(g_1)$ and $\text{orb}(g_2)$ for the conjugation action are either equal or are disjoint. State, and justify, the class equation for G. (7)

(d) Using the class equation, or otherwise, prove that if the cardinality of G equals p^r, where p is a prime number and $r \geq 1$, then the cardinality of the centre $Z(G)$ of G is not equal to 1. (6)

[25]

3. (a) Let G be a finite group of cardinality 56. Prove that G has a normal p-Sylow subgroup for some prime number p dividing 56. You may use, without proof, the Sylow theorems. (13)

(b) Let R be a ring which is an integral domain. Let a and b be elements of R and let $(a)_R$ and $(b)_R$ be the ideals of R generated by a and b respectively. Prove that

$$(a)_R = (b)_R$$

(an equality of ideals) if and only if

$$a = bu$$

where u is a unit of R. (12)

[25]

4. (a) Let $\mathbf{Z}$ be the ring of integers. Let R be the ring

$$\mathbf{Z}[\sqrt{11}] = \{a + b\sqrt{11} : a, b \in \mathbf{Z}\}$$

and let

$$I = (5, 1 + \sqrt{11})_R = \{5x + y(1 + \sqrt{11}) : x, y \in R\}$$

be the ideal of R generated by 5 and $1 + \sqrt{11}$.

By constructing a surjective ring homomorphism from R to a field F with kernel I, or otherwise, show that I is a maximal ideal of R. Clearly state any theorems from the lectures that you use. (12)

(b) Let R be a ring with **finitely** many elements. Assume that R is an integral domain and let a be a non zero element of R. Show that the map

$$\phi : R \to R$$

defined by

$$x \mapsto \phi(x) = ax$$

is a bijection. Deduce that R is a field. (8)

(c) Using the result in (b), or otherwise, prove that if R is a ring with finitely many elements and if I is a prime ideal of R, then I is a maximal ideal. (5)

[**25**]

[END OF PAPER]